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Studierichting: Informatica

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$$f(x) = \ln(x^2 + \sqrt{x^4 - 1})$$

$$\begin{aligned} f'(x) &= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(2x + \frac{1}{2\sqrt{x^4 - 1}} \cdot 4x^3 \right) \\ &= \frac{2x + \frac{2x^3}{\sqrt{x^4 - 1}}}{x^2 + \sqrt{x^4 - 1}} \\ &= \frac{\frac{2x\sqrt{x^4 - 1} + 2x^3}{\sqrt{x^4 - 1}}}{x^2 + \sqrt{x^4 - 1}} \\ &= \frac{2x(\sqrt{x^4 - 1} + x^2)}{\sqrt{x^4 - 1}(x^2 + \sqrt{x^4 - 1})} \\ &= \frac{2x}{\sqrt{x^4 - 1}} \end{aligned}$$

References